PART A

1. d)Overhead Triceps Extension

2. c) To assist the athlete if they struggle to complete the lift

3.b) Barbell Push Press

4.c) Quickness and change of direction

5. c) Supinated grip

6. d) To prevent dumbbells from dropping on chest or face

7. c) The rack position (shoulder height).

8.d) Exercises involving movements far from the body

9. b) During the hardest phase of the lift

10. d) Both B and C

PART B

11. Main steps involved in performing an EZ-bar curl.

- Stand with feet shoulder-width apart.
- Grasp the EZ-bar with a supinated (underhand) grip.
- Keep elbows close to your torso and back straight.
- Curl the weight upward by flexing your elbows, exhaling as you lift.
- Squeeze your biceps at the top of the movement.
- Lower the weight slowly back to the starting position, inhaling as you lower it.

12. The spotter should stand directly behind the lifter. They should place their hands close to the ends of the barbell, ready to grasp it. Alternatively, they can place their hands under the bar near the lifter's wrists, ready to assist the lift upward.

13. Joint structure and muscle tightness or flexibility.

14. The primary role is to be prepared to assist by helping to lift the barbell if the lifter fails or struggles to stand up from the bottom position, ensuring safety.

15. Pronated (overhand) grip and Hook grip.

16. Lie on a flat bench with your feet firmly on the floor. Hold a barbell or EZ-bar with a narrow overhand grip above your chest, arms fully extended. Keeping your upper arms stationery and elbows tucked in, lower the bar by flexing your elbows until the bar is near your forehead. Then, extend your elbows to return the bar to the starting position.

17. An alternated grip involves one hand using a pronated (overhand) grip and the other using a supinated (underhand) grip on the barbell. Its primary benefit is that it prevents the barbell from rolling out of the hands, allowing the lifter to hold onto heavier weights, most commonly during the deadlift.

18. The spotter should stand at the head of the bench. As the lifter struggles, the spotter should gently grasp the barbell with an alternated grip (for security) and provide just enough assistance to help the lifter complete the repetition. The assistance should be minimal, allowing the lifter to still exert effort. Once the repetition is complete, help the lifter rerack the barbell safely.

19. The hang shrug focuses on a powerful, vertical shrug of the shoulders to generate upward momentum on the barbell. The hang jump incorporates a full triple extension (ankles, knees, hips) and an explosive jump to drive the barbell upward with significantly more power, which is necessary to get under the bar for the catch.

20. The barbell bench press is a compound upper-body exercise where the lifter lies on a flat bench and lowers a weighted barbell to their chest and then presses it back up until their arms are fully extended. It primarily targets the pectoral muscles, along with the triceps and anterior deltoids.

21. Lifters use the Valsalva maneuver (exhaling against a closed glottis) to increase intra-abdominal and intra-thoracic pressure. This creates a rigid core, which stabilizes the spine and supports the torso under heavy load, reducing the risk of injury.

22. The spotter should stand at the head of the bench, directly behind the lifter, with a stable stance. They should be positioned close enough to grasp the barbell easily if the lifter needs assistance.

PART C

23.

- Highly Effective for Strength: Pull-ups are a closed-chain, compound bodyweight exercise that requires lifting one's entire body weight, building significant functional strength in the back, shoulders, and arms.
- Excellent for Muscle Hypertrophy: They are a primary movement for developing the latissimus dorsi (lats), which creates back width. They also heavily engage the biceps, rhomboids, traps, and core, promoting overall upper-body muscle growth.

- Versatility: By varying grip (wide, narrow, supinated, pronated), you can emphasize different muscle groups (e.g., a supinated grip targets the biceps more).
- Conclusion: Pull-ups are one of the most fundamental and effective exercises for building a strong, well-developed upper body.

24. Relationship: Good flexibility allows for a greater range of motion (ROM) around joints. This means muscles and tendons can lengthen more effectively during activity without reaching a point of strain or tear, thus reducing the risk of injuries like muscle pulls, strains, and joint sprains.

- Importance in Warm-up:
    - Increases Blood Flow: Dynamic stretching in a warm-up increases blood flow to muscles, making them more pliable and ready for work.
    - Improves Elasticity: It prepares the muscles and connective tissues for the demands of the workout, enhancing their elasticity and performance.
    - Primes Nervous System: It activates the nervous system, improving coordination and muscle recruitment.
    - Note: Static stretching (holding a stretch) is best saved for the cool-down, while dynamic stretching (movement-based) is ideal for warm-ups.

25.

- Argument for Alignment: Good body alignment ensures that forces are distributed correctly across joints and muscles during movement. Poor alignment (e.g., rounded back during a deadlift) places abnormal stress on specific structures (like spinal discs), leading to acute or chronic injuries, regardless of how strong or flexible you are.
- Counterpoint/Integration: Strength and flexibility are crucial supporting factors. Strength provides joint stability, and flexibility allows for proper alignment through a full ROM. A strong core, for example, is essential for maintaining spinal alignment.
- Evaluation: While all three are vital, good body alignment is the foundational priority. Without it, greater strength can sometimes lead to more severe injuries because you are moving heavy loads in a dysfunctional pattern. Flexibility without alignment control can lead to joint instability. Therefore, proper technique and alignment are the most critical elements for injury prevention, upon which strength and flexibility should be built.

26.

- Benefits:
    - Psychological: Provides confidence to attempt heavy lifts.
    - Assistance on Lockout: Can help "coax" the final part of the lift if the lifter stalls at the knees or during hip extension.

- Risks:
    - Safety Hazard: The conventional deadlift is extremely difficult to spot safely. A spotter standing in front risks injury if the lifter fails and drops the bar.
    - Interference: A spotter touching the bar prematurely can disrupt the lifter's balance and technique, potentially causing injury.
    - False Sense of Security: May encourage the lifter to attempt weights they are not truly ready for.
- Conclusion: For a conventional deadlift, the risks often outweigh the benefits. Safer alternatives include using bumper plates in a dedicated platform (so you can safely fail and drop the bar) or using rack pulls from pins. If a spotter is used, they should only assist at the top of the movement, not from the floor.

27. Barbell Push Press:
    - Start: Barbell on front rack position (resting on shoulders, elbows high).
    - Movement: A slight dip at the knees and hips (eccentric), followed by an explosive drive upward with the legs (concentric). As the leg drive decelerates, the arms and shoulders forcefully press the bar overhead to full arm extension.
    - Breathing: Inhale during the dip, exhale forcefully during the drive and press.
- Barbell Push Jerk:
    - Start: Identical to the push press (front rack).
    - Movement: A similar explosive leg drive. However, as the bar is propelled upward, the lifter rapidly re-dips their body by splitting the feet or bending the knees, "dropping" under the bar to catch it at full arm extension with straight elbows, with the body in a partial squat position. The lifter then stands up fully.
    - Breathing: Inhale during the initial dip, exhale during the drive. A breath may be held or quickly exhaled during the drop and catch under the bar.

28.

- Increased Demand: Single-leg exercises immediately reduce the base of support, placing a much higher demand on the body's balance systems (vestibular, visual, and proprioceptive) compared to bilateral exercises.
- Role of Balance: Balance is the ability to maintain the body's center of gravity over its base of support. In a single-leg squat or lunge, the stabilizer muscles of the ankle, knee, and hip (e.g., gluteus medius) must work intensely to prevent the pelvis from tilting and the body from falling sideways.
- Role of Coordination: Coordination is the ability to execute smooth, efficient, and controlled movements. It is required to synchronize the movement of the working leg (e.g., bending) with the stabilizing function of the core and supporting leg, all while maintaining an upright torso.

- Analysis: Balance and coordination are not just byproducts but are fundamental components of performing single-leg exercises correctly. A lack of either will result in poor form, compensatory movements (like the knee valgus), and an inability to generate meaningful force, ultimately increasing the risk of injury. These exercises are as much about training the nervous system for stability as they are about strengthening the muscles